FELIPE RIBEIRO

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EDUCATION

University at Buffalo, State University of New York — **Buffalo, NY**
Bachelor of Science in Mechanical Engineering — *Expected May 2027*
Relevant Coursework: Mechanics of Solids, Statics, Dynamics, Mechanics of Materials, Machines & Mechanisms, Finite Element Analysis, Control Systems, Manufacturing Processes, Thermodynamics, Heat Transfer

TECHNICAL SKILLS

Robotics & Embedded: Arduino, ESP32, Raspberry Pi, PID & Kalman filtering, Gimbal systems
Programming & Tools: C++, MATLAB, Python, MS Office (Excel, Word, PowerPoint)
CAD & Analysis: Fusion 360, SolidWorks, AutoCAD, Prusa Slicer, FEA

ENGINEERING EXPERIENCE

UB ASME — SDC Team ("PODIUM" Design Competition) — **Buffalo, NY**
Co-Lead, Payload Electronics & Automation Division — *June 2026 – Present*
- Co-leading a 12-person team responsible for all electronics and automation on the robot, for UB ASME's first year competing in the "PODIUM" Student Design Competition.
- Personally designing and building the payload systems for three of the competition's events — Lift, Cycle, and Marksman — from brainstorming the mechanisms to wiring the electronics.
- Handling the power system and sensors needed for the required 30-second autonomous run.
- Put together the budget request that got the team's funding raised from $500 to $750.

UB ASME — IAM3D 2026 Rover Team — **Buffalo, NY**
Electronics Lead — *August 2025 – May 2026*
- Lead the electronics for our 3D-printed rover, working with the mechanical, electrical, and software teams to get it ready to compete against other colleges.
- Designed and wired a power system with multiple batteries that boosted the rover's motor performance by about 25% while keeping voltage steady for the high-torque servos.
- Programmed the Arduino and ESP32 to take signals from the RC receiver and control the motors, servos, and relays.
- Competed with the team at the EFX competition at the University of Central Florida.

Four-Bar Linkage Prosthetic Hand — **Buffalo, NY**
Personal Project — *June 2026 – Present*
- Designing a prosthetic hand where each finger is driven by a servo and a fishing-line tendon through a four-bar linkage, letting it curl in and out.
- Modeled the finger movement in PMKS+ first to check the range of motion would work before building anything.
- Currently 3D-printing and assembling the finger parts and wiring up the servos and tendons.

Self-Balancing Inverted Pendulum Robot — **Buffalo, NY**
Personal Project — *February 2026*
- Built a self-balancing two-wheeled robot with an Arduino, MPU-6050 gyro/accelerometer, and DRV8833 motor driver, cutting its sway from +8°/−6° down to about ±4.5° by adding weight and mounting the sensor more securely.
- Combined the gyroscope and accelerometer readings with a Kalman filter and tuned a PID loop (Kp 90, Ki 0.2, Kd 6) to keep it balanced in real time.
- Noticed the two motors didn't start moving at the same point, so I tested each one and adjusted the code to account for the difference, which made the balancing noticeably smoother.

2-Axis Gimbal System — **Buffalo, NY**
Personal Project — *March 2026*
- Designed and 3D-printed a 2-axis camera gimbal that can be aimed remotely with servos.
- Added filtering and a deadband so the gimbal moves smoothly instead of twitching.
- Designed the frame to be light and balanced so the servos didn't have to work as hard.

Servo-Actuated 2-DOF Excavator Payload Mechanism — **Buffalo, NY**
Personal Project — *January 2026*
- Designed a 3D-printed excavator-style arm in Fusion 360 with two joints (elbow, wrist) to pick up and move objects.
- 3D-printed the bearings and gears myself and used high-torque servos at each joint to keep the movement smooth and cut down on friction.

PROFESSIONAL EXPERIENCE

Infusion by Castles — **New York, NY**
Advertising Intern — Client Operations & Office Design — *August 2022 – September 2024*
- Organized direct mail campaign files by market segment so clients could find what they needed, and helped redesign the office layout when the company relocated.